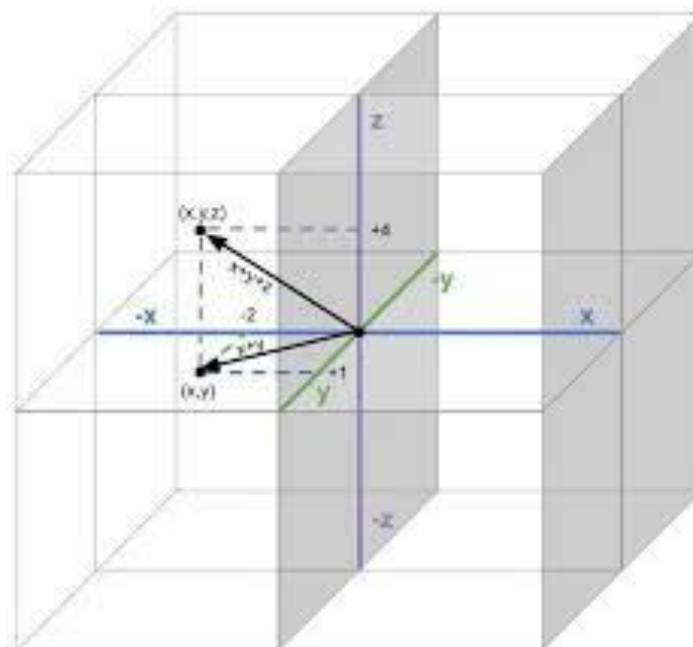


Linear Algebra

Math 1057

Instructor

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Lecture 13

Applications of the Determinant

13.1 The Cofactor Method

Recall that for $\mathbf{A} \in \mathbb{R}^{n \times n}$ we defined

$$\det \mathbf{A} = a_{j1}C_{j1} + a_{j2}C_{j2} + \cdots + a_{jn}C_{jn}$$

where $C_{jk} = (-1)^{j+k} \det \mathbf{A}_{jk}$ is called the (j, k) -Cofactor of $\mathbf{A}$ and

$$\mathbf{a}_j = [a_{j1} \quad a_{j2} \quad \cdots \quad a_{jn}]$$

is the j th row of $\mathbf{A}$. If $\mathbf{c}_j = [C_{j1} \quad C_{j2} \quad \cdots \quad C_{jn}]$ then

$$\det \mathbf{A} = [a_{j1} \quad a_{j2} \quad \cdots \quad a_{jn}] \begin{bmatrix} C_{j1} \\ C_{j2} \\ \vdots \\ C_{jn} \end{bmatrix} = \mathbf{a}_j \cdot \mathbf{c}_j^T.$$

Suppose that $\mathbf{B}$ is the matrix obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_j$ with a distinct row $\mathbf{a}_k$. To compute $\det \mathbf{B}$ expand along its j th row $\mathbf{b}_j = \mathbf{a}_k$:

$$\det \mathbf{B} = \mathbf{a}_k \cdot \mathbf{c}_j^T = 0.$$

The Cofactor Method is an alternative method to find the inverse of an invertible matrix. Recall that for any matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, if we expand along the j th row then

$$\det \mathbf{A} = \mathbf{a}_j \cdot \mathbf{c}_j^T.$$

On the other hand, if $j \neq k$ then

$$\mathbf{a}_j \cdot \mathbf{c}_k^T = 0.$$

In summary,

$$\mathbf{a}_j \cdot \mathbf{c}_k^T = \begin{cases} \det \mathbf{A}, & \text{if } j = k \\ 0, & \text{if } j \neq k. \end{cases}$$

Form the **Cofactor matrix**

$$\text{Cof}(\mathbf{A}) = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} = \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_n \end{bmatrix}.$$

Then,

$$\begin{aligned} \mathbf{A}(\text{Cof}(\mathbf{A}))^T &= \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \mathbf{c}_1^T & \mathbf{c}_2^T & \cdots & \mathbf{c}_n^T \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{a}_1 \mathbf{c}_1^T & \mathbf{a}_1 \mathbf{c}_2^T & \cdots & \mathbf{a}_1 \mathbf{c}_n^T \\ \mathbf{a}_2 \mathbf{c}_1^T & \mathbf{a}_2 \mathbf{c}_2^T & \cdots & \mathbf{a}_2 \mathbf{c}_n^T \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_n \mathbf{c}_1^T & \mathbf{a}_n \mathbf{c}_2^T & \cdots & \mathbf{a}_n \mathbf{c}_n^T \end{bmatrix} \\ &= \begin{bmatrix} \det \mathbf{A} & 0 & \cdots & 0 \\ 0 & \det \mathbf{A} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \det \mathbf{A} \end{bmatrix} \end{aligned}$$

This can be written succinctly as

$$\mathbf{A}(\text{Cof}(\mathbf{A}))^T = \det(\mathbf{A})\mathbf{I}_n.$$

Now if $\det \mathbf{A} \neq 0$ then we can divide by $\det \mathbf{A}$ to obtain

$$\mathbf{A} \left(\frac{1}{\det \mathbf{A}} \right) (\text{Cof}(\mathbf{A}))^T = \mathbf{I}_n.$$

This leads to the following formula for the inverse:

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} (\text{Cof}(\mathbf{A}))^T$$

Although this is an explicit and elegant formula for $\mathbf{A}^{-1}$, it is computationally intensive, even for 3×3 matrices. However, for the 2×2 case it provides a useful formula to compute

the matrix inverse. Indeed, if $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ we have $\text{Cof}(\mathbf{A}) = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$ and therefore

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

When does an integer matrix have an integer inverse? We can answer this question using the Cofactor Method. Let us first be clear about what we mean by an integer matrix.

Definition 13.1: A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is called an **integer matrix** if every entry of $\mathbf{A}$ is an integer.

Suppose that $\mathbf{A} \in \mathbb{R}^{n \times n}$ is an invertible integer matrix. Then $\det(\mathbf{A})$ is a non-zero integer and $(\text{Cof}(\mathbf{A}))^T$ is an integer matrix. If $\mathbf{A}^{-1}$ is also an integer matrix then $\det(\mathbf{A}^{-1})$ is also an integer. Now $\det(\mathbf{A}) \det(\mathbf{A}^{-1}) = 1$ thus it must be the case that $\det(\mathbf{A}) = \pm 1$. Suppose on the other hand that $\det(\mathbf{A}) = \pm 1$. Then by the Cofactor method

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} (\text{Cof}(\mathbf{A}))^T = \pm (\text{Cof}(\mathbf{A}))^T$$

and therefore $\mathbf{A}^{-1}$ is also an integer matrix. We have proved the following.

Theorem 13.2: An invertible integer matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has an integer inverse $\mathbf{A}^{-1}$ if and only if $\det \mathbf{A} = \pm 1$.

We can use the previous theorem to generate integer matrices with an integer inverse as follows. Begin with an upper triangular matrix $\mathbf{M}_0$ having integer entries and whose diagonal entries are either 1 or -1 . By construction, $\det(\mathbf{M}_0) = \pm 1$. Perform any sequence of elementary row operations of Type 1 and Type 3. This generates a sequence of matrices $\mathbf{M}_1, \dots, \mathbf{M}_p$ whose entries are integers. Moreover,

$$\mathbf{M}_0 \sim \mathbf{M}_1 \sim \dots \sim \mathbf{M}_p.$$

Therefore,

$$\pm 1 = \det(\mathbf{M}) = \det(\mathbf{M}_1) = \dots = \det(\mathbf{M}_p).$$

13.2 Cramer's Rule

The Cofactor method can be used to give an explicit formula for the solution of a linear system where the coefficient matrix is invertible. The formula is known as Cramer's Rule. To derive this formula, recall that if $\mathbf{A}$ is invertible then the solution to $\mathbf{Ax} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. Using the Cofactor method, $\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}(\text{Cof}(\mathbf{A}))^T$, and therefore

$$\mathbf{x} = \frac{1}{\det \mathbf{A}} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Consider the first component x_1 of $\mathbf{x}$:

$$x_1 = \frac{1}{\det \mathbf{A}}(b_1 C_{11} + b_2 C_{21} + \cdots + b_n C_{n1}).$$

The expression $b_1 C_{11} + b_2 C_{21} + \cdots + b_n C_{n1}$ is the expansion of the determinant along the **first** column of the matrix obtained from $\mathbf{A}$ by replacing the first column with $\mathbf{b}$:

$$\det \begin{bmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ b_2 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{bmatrix} = b_1 C_{11} + b_2 C_{21} + \cdots + b_n C_{n1}$$

Similarly,

$$x_2 = \frac{1}{\det \mathbf{A}}(b_1 C_{12} + b_2 C_{22} + \cdots + b_n C_{n2})$$

and $(b_1 C_{12} + b_2 C_{22} + \cdots + b_n C_{n2})$ is the expansion of the determinant along the **second** column of the matrix obtained from $\mathbf{A}$ by replacing the second column with $\mathbf{b}$. In summary:

Theorem 13.3: (Cramer's Rule) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be an invertible matrix. Let $\mathbf{b} \in \mathbb{R}^n$ and let $\mathbf{A}_i$ be the matrix obtained from $\mathbf{A}$ by replacing the i th column with $\mathbf{b}$. Then the solution to $\mathbf{Ax} = \mathbf{b}$ is

$$\mathbf{x} = \frac{1}{\det \mathbf{A}} \begin{bmatrix} \det \mathbf{A}_1 \\ \det \mathbf{A}_2 \\ \vdots \\ \det \mathbf{A}_n \end{bmatrix}.$$

Although this is an explicit and elegant formula for $\mathbf{x}$, it is computationally intensive, and used mainly for theoretical purposes.

13.3 Volumes

The volume of the parallelepiped determined by the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is

$$\text{Vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \text{abs}(\mathbf{v}_1^T(\mathbf{v}_2 \times \mathbf{v}_3)) = \text{abs}(\det [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3])$$

where $\text{abs}(x)$ denotes the absolute value of the number x . Let $\mathbf{A}$ be an invertible matrix and let $\mathbf{w}_1 = \mathbf{A}\mathbf{v}_1, \mathbf{w}_2 = \mathbf{A}\mathbf{v}_2, \mathbf{w}_3 = \mathbf{A}\mathbf{v}_3$. How are $\text{Vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ and $\text{Vol}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$ related? Compute:

$$\begin{aligned} \text{Vol}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3) &= \text{abs}(\det [\mathbf{w}_1 \ \mathbf{w}_2 \ \mathbf{w}_3]) \\ &= \text{abs}(\det [\mathbf{A}\mathbf{v}_1 \ \mathbf{A}\mathbf{v}_2 \ \mathbf{A}\mathbf{v}_3]) \\ &= \text{abs}(\det(\mathbf{A} [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3])) \\ &= \text{abs}(\det \mathbf{A} \cdot \det [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]) \\ &= \text{abs}(\det \mathbf{A}) \cdot \text{Vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3). \end{aligned}$$

Therefore, the number $\text{abs}(\det \mathbf{A})$ is the factor by which volume is changed under the linear transformation with matrix $\mathbf{A}$. In summary:

Theorem 13.4: Suppose that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are vectors in $\mathbb{R}^3$ that determine a parallelepiped of non-zero volume. Let $\mathbf{A}$ be the matrix of a linear transformation and let $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ be the images of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ under $\mathbf{A}$, respectively. Then

$$\text{Vol}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3) = \text{abs}(\det \mathbf{A}) \cdot \text{Vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3).$$

Example 13.5. Consider the data

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}, \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 5 \\ 1 \end{bmatrix}.$$

and let $\mathbf{w}_1 = \mathbf{A}\mathbf{v}_1, \mathbf{w}_2 = \mathbf{A}\mathbf{v}_2$, and $\mathbf{w}_3 = \mathbf{A}\mathbf{v}_3$. Find the volume of the parallelepiped spanned by the vectors $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$.

Solution. We compute:

$$\text{Vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \text{abs}(\det([\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3])) = \text{abs}(-7) = 7$$

We compute:

$$\det(\mathbf{A}) = 55.$$

Therefore, the volume of the parallelepiped spanned by the vectors $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is

$$\text{Vol}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3) = \text{abs}(55) \times 7 = 385.$$

□

After this lecture you should know the following:

- what the Cofactor Method is
- what Cramer's Rule is
- the geometric interpretation of the determinant (volume)